

Model selection

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Warmup

Work on the warmup activity (handout).

Warmup

Which model would you prefer?

```
m_dengue1 <- glm(Dengue ~ Age + WBC + PLT,  
                  data = dengue,  
                  family = binomial)  
m_dengue1$deviance
```

```
## [1] 5200.823
```

```
m_dengue2 <- glm(Dengue ~ Age + WBC + PLT + HCT,  
                  data = dengue, family = binomial)  
m_dengue2$deviance
```

```
## [1] 5199.946
```

Another example

```
m_dengue1 <- glm(Dengue ~ Age + WBC + PLT,  
                  data = dengue,  
                  family = binomial)
```

```
m_dengue1$aic
```

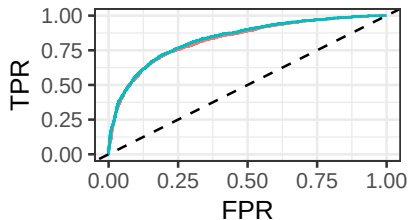
```
## [1] 5208.823
```

```
m_dengue2 <- glm(Dengue ~ Age + WBC + PLT +  
                  BMI + Temperature + Vomiting,  
                  data = dengue, family = binomial)
```

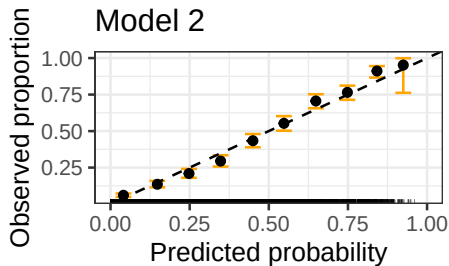
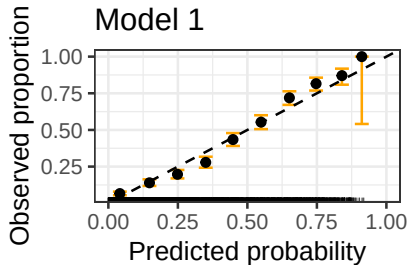
```
m_dengue2$aic
```

```
## [1] 5107.318
```

Another example



— Model 1 — Model 2



Key points

- ▶ Deviance (on the training data) can **never** increase when I add additional explanatory variables – don't use raw deviance to compare models with different numbers of predictors
- ▶ AIC is a better method to compare models (estimate of predictive ability on new data)
- ▶ But, a difference in AIC does not always have **practical significance**
- ▶ Other criteria for choosing a model: type of research question, model interpretability, model parsimony, ease of use by practitioners, model diagnostics

Types of research question

For what different types of research question might I use a regression model?

Other ways of comparing models

Alternatives to AIC:

LRT:

Activity

Work on the class activity (handout).

Activity

LRT:

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

Reject when $G > \chi^2_{q,\alpha}$.

Activity

- ▶ LRT will choose the full model if $G > \chi_{q,\alpha}^2$
- ▶ AIC will choose the full model if $G > 2q$
- ▶ LRT is more strict when $\chi_{q,\alpha}^2 > 2q$

```
alpha = 0.05
```

```
q <- 1:10
```

```
qchisq(1 - alpha, q) - 2*q
```

```
## [1] 1.84145882 1.99146455 1.81472790 1.48772904 1.19026455 0.95424273 0.73675727 0.53778982 0.35523980 0.19151043
## [7] 0.06714045 -0.49268694 -1.08102240 -1.69296195 -2.31973805 -2.95026195 -3.58023980 -4.20975727 -4.83825727 -5.46573982
```

```
alpha = 0.01
```

```
q <- 1:20
```

```
qchisq(1 - alpha, q) - 2*q
```

```
## [1] 4.634897e+00 5.210340e+00 5.344867e+00 5.276704e+00 5.000000e+00 4.518139e+00 3.937838e+00 3.268289e+00 2.529435e+00 1.741573e+00 9.151458e-01 3.223747e-01 -2.893281e-01 -5.542551e-01 -8.182821e-01 -1.082318e+00 -1.346355e+00 -1.610392e+00 -1.874430e+00 -2.138467e+00
## [6] 4.811894e+00 4.475307e+00 4.090235e+00 3.665994e+00 3.202761e+00 2.700528e+00 2.169295e+00 1.609062e+00 1.030829e+00 5.346596e-01 1.234567e-01 -3.456789e-01 -7.890123e-01 -1.234567e+00 -1.678901e+00 -2.123456e+00 -2.567890e+00 -3.012345e+00 -3.456789e+00 -3.901234e+00
## [11] 2.724970e+00 2.216967e+00 1.688250e+00 1.141238e+00 6.789012e-01 2.123456e-01 -2.345678e-01 -4.567890e-01 -6.789012e-01 -8.901234e-01 -1.101234e+00 -1.312345e+00 -1.523456e+00 -1.734567e+00 -1.945678e+00 -2.156789e+00 -2.367890e+00 -2.578901e+00 -2.789012e+00 -2.990123e+00
## [16] -7.309118e-05 -5.913364e-01 -1.194694e+00 -1.809133e+00 -2.418266e+00 -3.027399e+00 -3.636532e+00 -4.245665e+00 -4.854798e+00 -5.463931e+00 -6.073064e+00 -6.682197e+00 -7.291330e+00 -7.900463e+00 -8.509596e+00 -9.118729e+00
```

AIC vs. LRT

- ▶ LRT can only be used with nested models; AIC can compare non-nested models
- ▶ LRT is more stringent when q (difference in number of parameters between models) is small
- ▶ AIC is more stringent when q is large
- ▶ But, they ask different questions!
 - ▶ LRT: Is difference in model deviance due to chance?
 - ▶ AIC: Estimate expected prediction error
- ▶ **Neither** is asking whether difference in models is **meaningful** (*practical significance*)